Ahmedabad University

CSE400 Fundamentals of Probability in Computer Science

Section -1

Computer Networks

Group 16

Milestone 3

Optimizing UAV deployment for Multi-UAV Cooperative Sensing

Project Context and Motivation

- Project: Eigenvalue-Based Cooperative UAV Target Detection
- Domain: UAV surveillance & wireless sensing

Why uncertainty exists in the system:

- Targets are found in a surveillance region by several UAVs.
- Environmental noise has an impact on sensors
- Binary hypothesis testing is used to simulate detection.

  H_0: Only noise

  H_1: Target signal with noise
- Objective: High Pd, Low Pfa.

Project Objective and Deterministic Baseline

Goal: Collaborative UAV target identification in the face of uncertainty

- Deterministic Foundation:

  5000 x 5000 m is the surveillance area.

  $0 \leq X \leq 1800$ is the deployable region.

  Rule for UAV placement: $X = 900$

  Equal spacing along the Y-axis

  $M = 3$, 4, and 5 UAV scenarios

  Offers well distributed baseline coverage.

System Pipeline Overview

- Initialization of the system parameters
- Generation of the target position
- Deployment of deterministic UAVs
- Distance computation
- Angle of the antenna
- Signal detection using noise
- Detection probability calculation
- Analyzing performance

Noise, signal fluctuation, and target location are all sources of uncertainty.

Sources of Uncertainty

- Environmental noise:

  Gaussian noise in the sensors

- Target signal probability:

  Signal impacted by channel circumstances

- The target location:

  Targets can be seen in several places.

- Creates confusion in the signals that are received and detected

Key Random Variables

- Noise: $N \sim \text{Normal}(0, \sigma^2)$
- Target Signal: $S \sim \text{Normal}(0, \sigma s^2)$
- Target Position:

  $Xt \sim \text{Uniform}(1800,5000)$

  $Yt \sim \text{Uniform}(0,5000)$
- Observed Signal:

  $Y = S + N$ (target present)

Probabilistic Modeling

- AWGN-modeled noise
- Detection relies on SNR
- Eigenvalue-based model for detection

Detection Probability:

$$P_d = Q\left(Q^{-1}(P_{fa}) - \frac{SNR\sqrt{K/2}}{1+SNR}\right)$$

Cooperative Detection:

$$P_n = 1 - \prod_{m=1}^{M}(1 - P_{m,n})$$

Total Probability:

Psum = ΣPn

Deterministic Baseline Logic

- Fixed rules are used for UAV positioning.
- Every UAV was positioned at X = 900.
- uniform distance along the Y-axis
- Determine the distance to each target.
- Choose the closest target.
- Point the antenna in the direction of the target.
- Conduct detection and sensing

Simulation Implementation

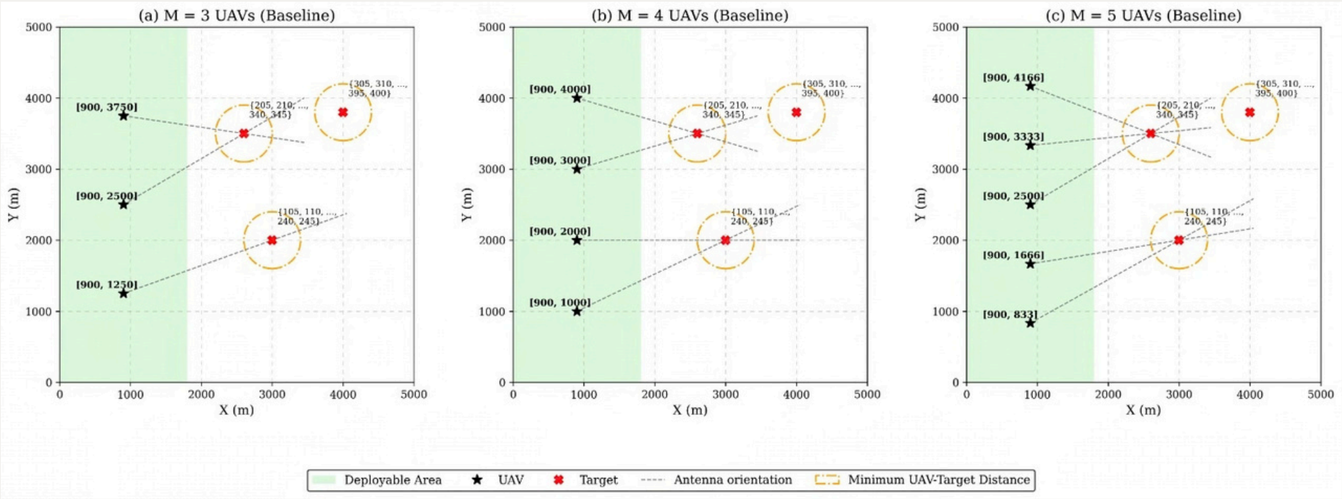
Techniques:

- UAV placement that is deterministic
- The computation of Euclidean distance
- The closest target choice
- Calculating the chance of detection

Pipeline:

- Input of parameters
- UAV deployment
- Simulated detection of targets
- assessment of performance

Results and Performance Analysis



Metrics

- Target distances for UAVs
- The likelihood of each target being detected
- Sum Detection Probability

Observations

- More UAVs equate to improved coverage
- Closer targets result in increased detection
- Equilibrium coverage and deterministic positioning

Limitations of Deterministic Strategy

- UAV location is fixed.
- Few configurations were explored.
- The likelihood of detection might not be optimized.
- Inability to adjust to various target distributions.

Summary & Transition

Key Takeaways:

- Deployment that is deterministic offers a baseline
- The impact of distance and noise on performance
- Coverage is improved by increasing UAVs.

Next Step:

- Employ random algorithms.
- Optimize the deployment of UAVs
- Enhance the performance of detecting